

Experiment 1: Hotel Reservation and Booking System

Question

Develop a **Hotel Reservation and Booking System** using **Python and SQLite**. The system should maintain customer information, room details, reservations, and payments. It should demonstrate customer management, room booking, reservation updates, and invoice generation while maintaining a normalized database structure.

Aim:

To design and implement a **Hotel Reservation and Booking System** using **Python and SQLite** for managing customers, rooms, bookings, payments, reservation updates, and invoice generation.

Program:

```
import sqlite3

# Connect Database
conn = sqlite3.connect("hotel.db")
cur = conn.cursor()

# Create Tables
cur.execute("CREATE TABLE IF NOT EXISTS Customer(ID INTEGER, Name TEXT)")
cur.execute("CREATE TABLE IF NOT EXISTS Room(RoomNo INTEGER, Type TEXT, Price INTEGER, Status TEXT)")
cur.execute("CREATE TABLE IF NOT EXISTS Booking(ID INTEGER, CustomerID INTEGER, RoomNo INTEGER, CheckIn TEXT, CheckOut TEXT)")
cur.execute("CREATE TABLE IF NOT EXISTS Payment(ID INTEGER, BookingID INTEGER, Amount INTEGER)")

# Clear Old Data
cur.execute("DELETE FROM Customer")
cur.execute("DELETE FROM Room")
cur.execute("DELETE FROM Booking")
```

```
cur.execute("DELETE FROM Payment")
```

```
# Insert Data
```

```
cur.execute("INSERT INTO Customer VALUES(1,'Rahul')")
```

```
cur.executemany("INSERT INTO Room VALUES(?,?,?,?)",[
```

```
(101,"Standard",2000,"Available"),
```

```
(102,"Deluxe",3000,"Available"),
```

```
(103,"Suite",5000,"Available")
```

```
])
```

```
# Book Room
```

```
cur.execute("INSERT INTO Booking VALUES(1,1,101,'2026-08-01','2026-08-03')")
```

```
cur.execute("UPDATE Room SET Status='Booked' WHERE RoomNo=101")
```

```
cur.execute("INSERT INTO Payment VALUES(1,1,6000)")
```

```
conn.commit()
```

```
# Show Booking
```

```
print("Booking Details")
```

```
cur.execute("""
```

```
SELECT Name, RoomNo, CheckIn, CheckOut
```

```
FROM Customer, Booking
```

```
WHERE Customer.ID = Booking.CustomerID
```

```
""")
```

```
for row in cur.fetchall():
```

```
    print(row)
```

```
# Update Booking
```

```
cur.execute("UPDATE Booking SET CheckOut='2026-08-04' WHERE ID=1")
```

```
conn.commit()
```

```
print("\nUpdated Booking")
```

```

cur.execute("SELECT * FROM Booking")

print(cur.fetchall())


# Show Invoice

print("\nInvoice")

cur.execute("""
SELECT Customer.Name, Payment.Amount
FROM Customer, Booking, Payment
WHERE Customer.ID=Booking.CustomerID
AND Booking.ID=Payment.BookingID
""")

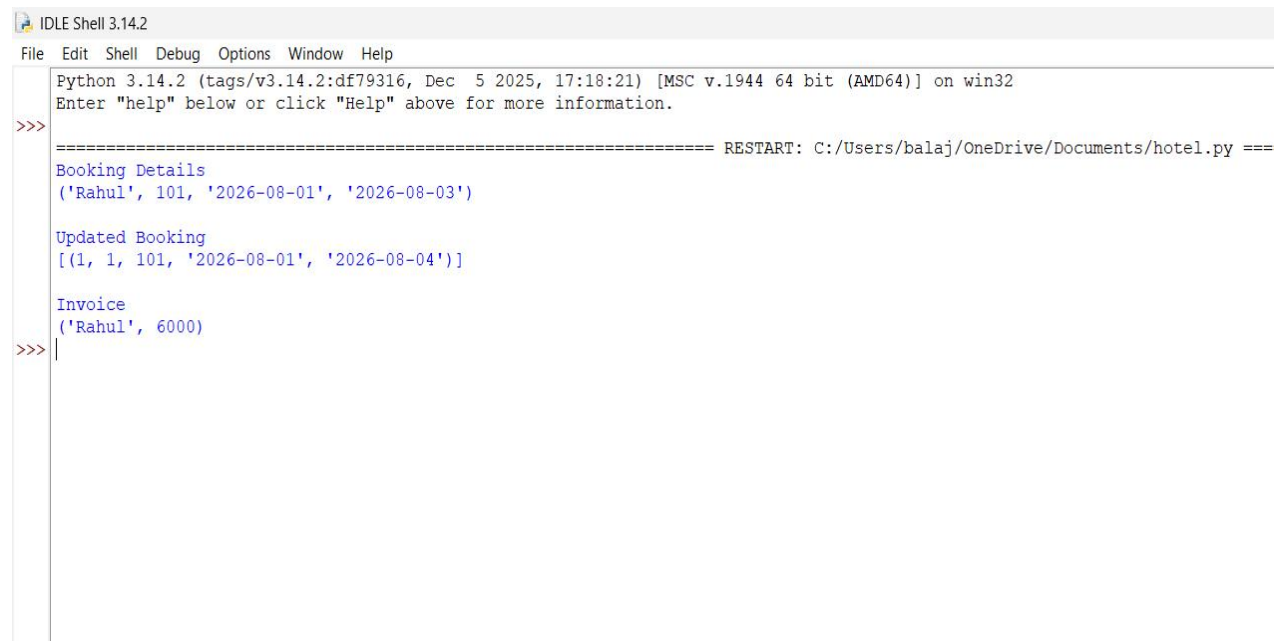
for row in cur.fetchall():

    print(row)


conn.close()

```

OUTPUT:



```

IDLE Shell 3.14.2
File Edit Shell Debug Options Window Help
Python 3.14.2 (tags/v3.14.2:df79316, Dec 5 2025, 17:18:21) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>> ===== RESTART: C:/Users/balaj/OneDrive/Documents/hotel.py =====
Booking Details
('Rahul', 101, '2026-08-01', '2026-08-03')

Updated Booking
[(1, 1, 101, '2026-08-01', '2026-08-04')]

Invoice
('Rahul', 6000)
>>>

```

Experiment 2: Employee Payroll Management System

Question

Develop an Employee Payroll Management System using Python and SQLite. The system should maintain employee details, department information, attendance records, salary details, and payroll information. It should demonstrate employee management, payroll generation, salary updates, and payroll display using database connectivity.

Aim

To design and implement an Employee Payroll Management System using Python and SQLite for managing employee details, departments, attendance, salary, and payroll generation.

Program

```
import mysql.connector

# Connect to MySQL

conn = mysql.connector.connect(
    host="localhost",
    user="root",
    password="kirth123",
    database="employee_payroll"
)

cur = conn.cursor()

# Create Tables

cur.execute("""
CREATE TABLE IF NOT EXISTS Employee(
    ID INT PRIMARY KEY,
    Name VARCHAR(50),
    Salary INT
)
""")

cur.execute("""
```

```
CREATE TABLE IF NOT EXISTS Payroll(  
    ID INT PRIMARY KEY,  
    Tax INT,  
    NetSalary INT  
)  
""")
```

```
# Delete Old Data
```

```
cur.execute("DELETE FROM Employee")
```

```
cur.execute("DELETE FROM Payroll")
```

```
# Insert Employee
```

```
cur.execute("INSERT INTO Employee VALUES(1,'Rahul',30000)")
```

```
# Calculate Payroll
```

```
salary = 30000
```

```
tax = 3000
```

```
net = salary - tax
```

```
cur.execute("INSERT INTO Payroll VALUES(%s,%s,%s)", (1, tax, net))
```

```
conn.commit()
```

```
# Display Employee Details
```

```
print("Employee Details")
```

```
cur.execute("SELECT * FROM Employee")
```

```
for row in cur.fetchall():
```

```
    print(row)
```

```
# Update Salary
```

```
cur.execute("UPDATE Employee SET Salary=32000 WHERE ID=1")
```

```
cur.execute("UPDATE Payroll SET Tax=3200, NetSalary=28800 WHERE ID=1")
conn.commit()
```

```
# Display Payroll
```

```
print("\nPayroll Details")
```

```
cur.execute("""
```

```
SELECT Employee.Name, Employee.Salary, Payroll.Tax, Payroll.NetSalary
```

```
FROM Employee
```

```
JOIN Payroll
```

```
ON Employee.ID = Payroll.ID
```

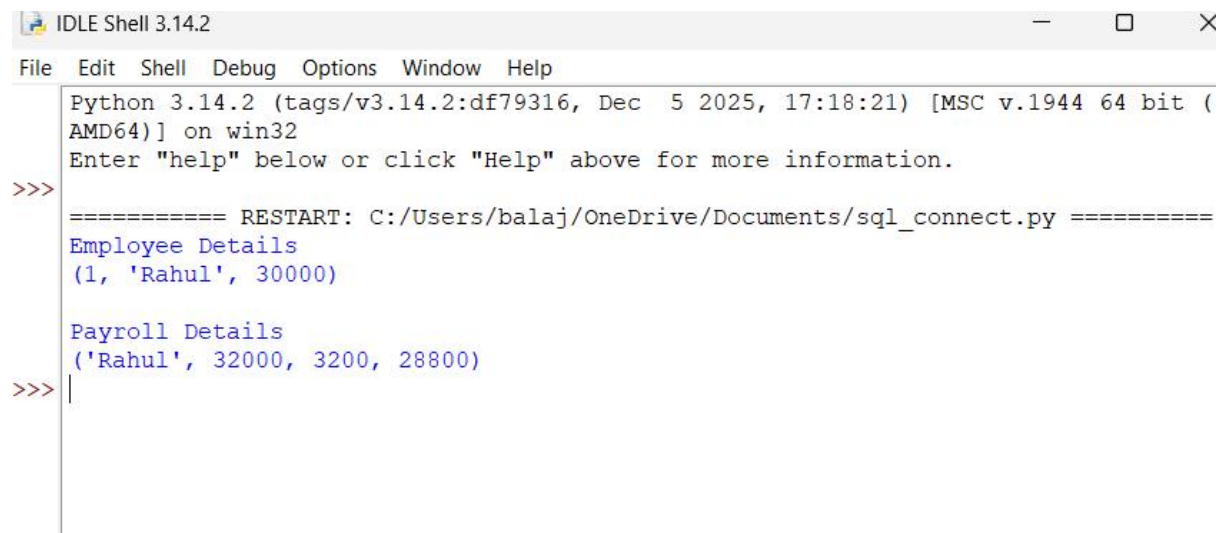
```
""")
```

```
for row in cur.fetchall():
```

```
    print(row)
```

```
conn.close()
```

OUTPUT:



```
IDLE Shell 3.14.2
File Edit Shell Debug Options Window Help
Python 3.14.2 (tags/v3.14.2:df79316, Dec 5 2025, 17:18:21) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:/Users/balaj/OneDrive/Documents/sql_connect.py =====
Employee Details
(1, 'Rahul', 30000)

Payroll Details
('Rahul', 32000, 3200, 28800)
>>> |
```